

Math 2263 Group Quiz 1

Names: _____

1. Either find the following limits or show they do not exist:

a. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\ln(x^2 + y^2)}$

b. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin(x/y)}{x^2 + y^2}$

c. $\lim_{(x,y) \rightarrow (0,0)} \frac{x(x^2 + y^2)}{y^2 + (x^2 + y^2)^2}$

2. If $z(x, y) = x^2y + x$, $x = r \cos \vartheta$, and $y = r \sin \vartheta$, find $\frac{\partial^2 z}{\partial r \partial \vartheta}$.

3. Let $f(x, y) = y^2 e^{x^2 - 2xy}$. Find the x, y values where $\nabla f = \mathbf{0}$. What does $\nabla f = \mathbf{0}$ mean geometrically?